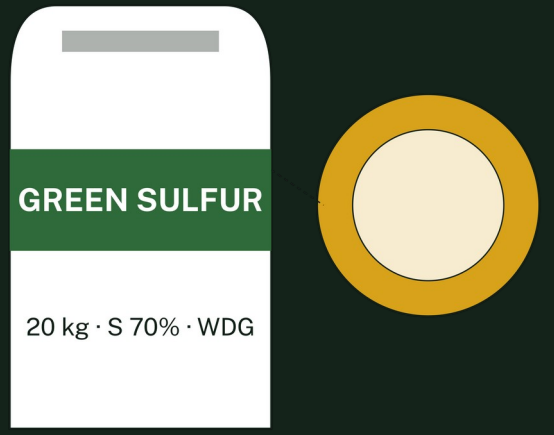
DEALER TRAINING · NOVEMBER 2026

Green Sulfur



Elemental sulfur 70%: the granule (WDG) for the soil, the powder for fertigation; 1, 5 and 20 kg, made in Lahore.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

Green Sulfur is elemental sulfur, 70%, in 2 grades: the granule (WDG), spread on the soil, and the fertigation powder, put through the water. The crop takes up sulfur only as sulfate, which moves with the water like nitrate and can wash away before the crop needs it. Elemental sulfur stays in the soil until microbes turn it into sulfate during the season; they work faster in warm, moist soil and fastest on fine particles. The granule breaks into fine particles in water. The plant needs sulfur to turn applied nitrogen into protein. The Punjab soil survey did not test sulfur. Wheat takes 1 pack of 1 kg at germination and 1 at early growth through the water; maize takes 5 kg drilled at sowing and 5 kg at germination, in 1 kg packs. Apply it before the crop needs it, because the microbes need time.

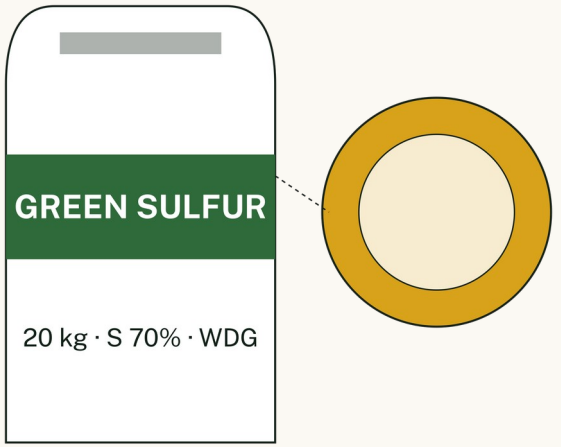
The 4 points it must cover

1. The bag: elemental sulfur 70%, PS 5449-2020; the granule (WDG) for the soil, the powder for the water; 1, 5 and 20 kg. 14 kg of sulfur in a 20 kg bag.
2. The nutrient: the crop takes up sulfur only as sulfate. Sulfate leaches like nitrate; elemental sulfur stays until microbes turn it into sulfate over the season.
3. The granule: it handles as a dry granule and breaks into fine particles in water. Microbes work on the surface of fine particles, so particle size decides how fast sulfate forms.
4. The method: drilled at sowing, or broadcast and worked in if there is no drill; through the water where the crop plan says. Apply it before the crop needs it.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the bag, in the published crop plan or in the named evidence on the slides that follow.

THE BAG

What the bag says, and what it is registered as.



20 kg

the bag. It holds 14 kg of elemental sulfur by the declared analysis. The 5 kg pack holds 3.5 kg and the 1 kg pack 0.7 kg.

S 70%

elemental sulfur, minimum, as the label declares it. The granule also contains a wetting agent (3 to 6%) and a dispersing agent (5 to 10%), which make it disperse in water.

WDG

the soil grade, a water-dispersible granule: it breaks into fine particles when it meets water. Fertigation uses the powder grade. Both grades come in 1, 5 and 20 kg.

3 packs

Rs 950 for 1 kg, Rs 4,250 for 5 kg and Rs 15,750 for 20 kg: the delivered farmer price, VAN farmer price sheet, August 2026.

0

of the 770,160 samples in the Punjab soil survey were tested for sulfur. A soil survey that did not test for sulfur cannot show it as deficient.

LAB 336

every batch is released through VAN Lab, accredited by the Pakistan National Accreditation Council (PNAC), before it ships.

Registered as	Pakistan Standard	Manufacturing licence
Green Sulfur	PS 5449-2020	CM/L-2962/2023

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“My wheat is pale. I have already given it 2 bags of urea and the second bag did nothing. What is it short of?”

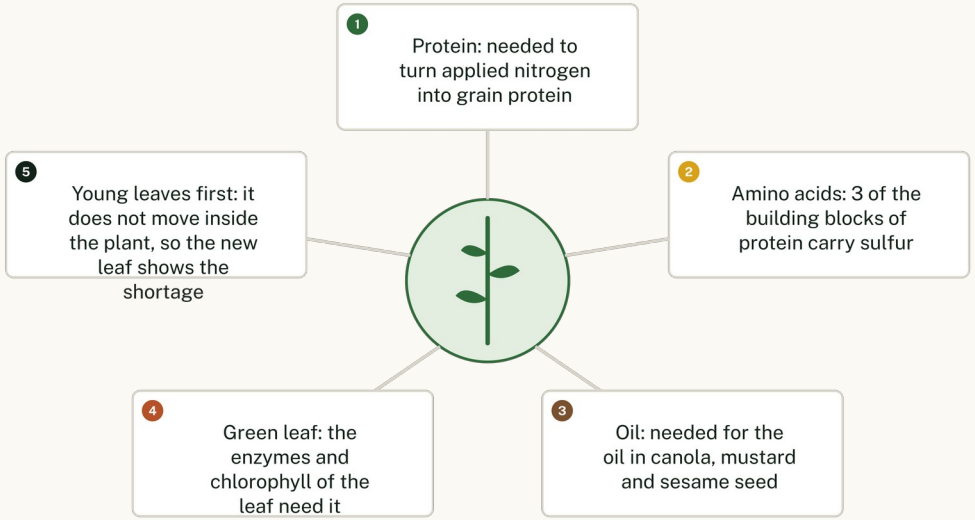
The most useful field check a dealer can make is which leaves turned pale first. A pale crop that does not respond to nitrogen is the most common case of sulfur shortage a dealer meets, and it is usually mistaken for nitrogen shortage. The answer has 3 parts: sulfur and protein, sulfate that washes away, and sulfate from the granule later.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

The plant needs sulfur to turn nitrogen into protein. Check the young leaves.

Sulfur is in 3 of the amino acids from which the plant builds protein. Without enough sulfur, the plant cannot turn the nitrogen it takes up into protein, so the extra urea is wasted and the crop stays pale. Sulfur shortage and nitrogen shortage look alike from a distance; the way to tell them apart is where the yellowing starts. Nitrogen moves inside the plant, so a nitrogen shortage turns the old leaves pale first. Sulfur does not move, so a sulfur shortage turns the young leaves pale first, and the whole young leaf is pale. If the yellowing did not respond to urea, look at the youngest leaf.



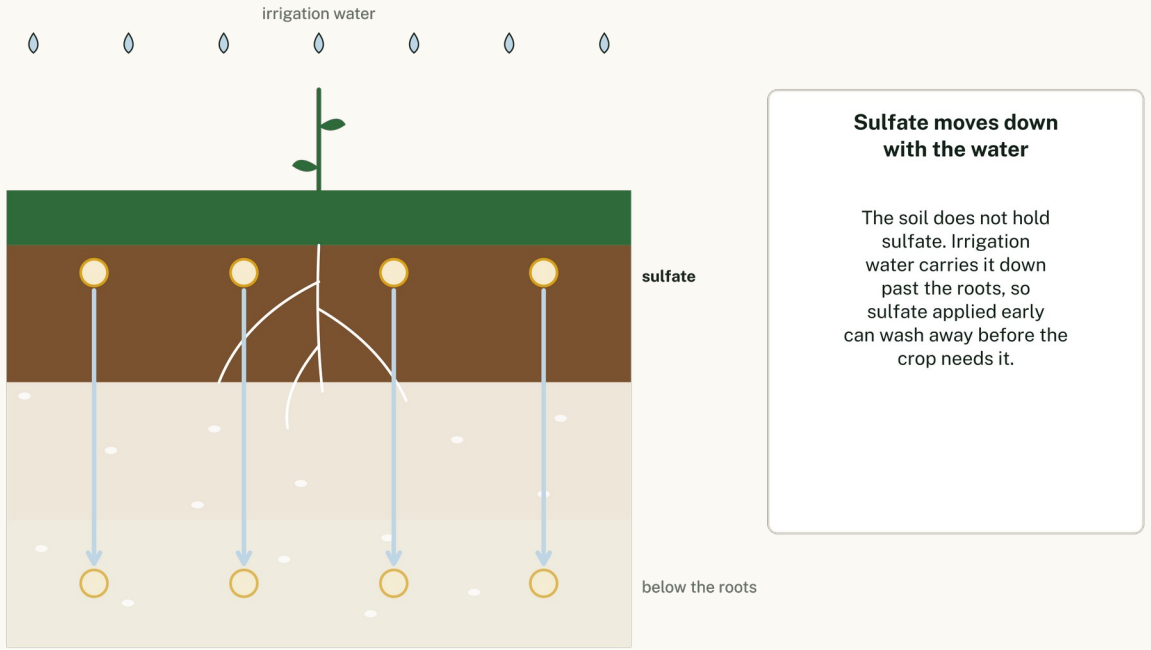
The jobs of sulfur, from the 17-nutrients page on van.com.pk. The youngest-leaf test is general agronomy, and a dealer can do it at the edge of the field.

Say it like this: “If the new leaves are the pale ones and the urea did nothing, the crop is short of sulfur.”

ANSWER 2 OF 3

The crop takes up sulfur only as sulfate, and sulfate moves with the water.

The root takes up sulfur in 1 form, sulfate. Sulfate carries a negative charge, like nitrate, so the soil does not hold it. It moves down with the irrigation water, past the root. On irrigated land, sulfate applied at sowing can be washed below the root before the crop needs it. Elemental sulfur is different. The plant cannot take it up as applied. Soil microbes oxidise it to sulfate a little at a time, and while it is still elemental it does not leach. Neither form is better in itself: a sulfate source corrects a shortage at once, and elemental sulfur supplies sulfate through the season.



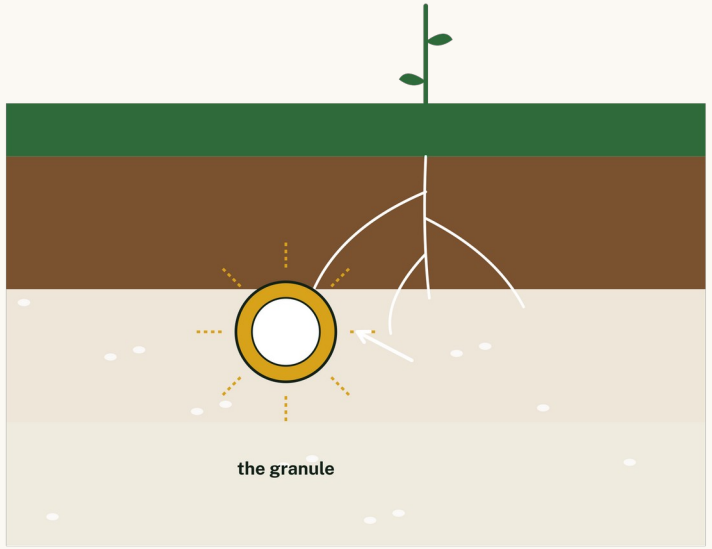
The drawing shows sulfate moving with the water. No leaching percentage is given, because VAN has no measurement for Pakistani fields.

Say it like this: “Sulfate moves down with the water. This sulfur stays in the soil until microbes turn it into sulfate, a little at a time.”

ANSWER 3 OF 3

The granule breaks into fine particles, which the soil microbes can work on.

Oxidation happens on the surface of the particle. The microbes can work only on the sulfur they can touch, so the surface area, not the quantity, sets how fast sulfate forms. Fine sulfur powder cannot be spread: it drifts and cakes. The Green Sulfur granule, the soil grade, is a water-dispersible granule: 70% sulfur with a wetting agent and a dispersing agent, so it stores and spreads as a granule and breaks into fine particles as soon as it meets water. While the microbes work, the soil around each particle becomes acidic, which can bring phosphate and zinc held by lime back into solution there. The soil becomes acidic only within a few millimetres of each particle, and only for the weeks the conversion takes. The acidity does not change the pH of the field.



70% sulfur, held as a granule. Water breaks it into fine particles. Microbes turn them into sulfate over the season, and the soil within a few millimetres of each one becomes acidic.

The granule in the root zone, with the small acidic zone around the particles. No particle size or oxidation rate is given.

Say it like this: “You spread it as a dry granule. In water it breaks into fine particles that the soil microbes can work on.”

THE EVIDENCE

What is verified for Green Sulfur, and what is not.

No trial of Green Sulfur itself is on VAN's record. A dealer may state the following as facts, and should say where each comes from: what the bag declares, the licence, the Punjab soil survey, the published chemistry and the youngest-leaf field test below.

Item	What it says	How a dealer may use it
The bag	Green Sulfur, elemental sulfur 70% minimum, in 2 grades: the granule (WDG) for the soil, with a wetting agent 3 to 6% and a dispersing agent 5 to 10%, and the fertigation powder for the water. Both grades come in packs of 1 kg, 5 kg and 20 kg.	As fact. The 70% is the sulfur figure; the agents make it a water-dispersible granule.
The licence	Pakistan Standard PS 5449-2020, manufacturing licence CM/L-2962/2023.	As fact. The number, never an expiry date.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	Sulfur was tested in 0 of them. It appears in no district file and no column, so it is never reported deficient and never enters a fertiliser plan built on the report.	As fact about the record. Never as "every field is short of sulfur".
The chemistry (published)	Sulfate leaches like nitrate. Elemental sulfur is oxidised by soil microbes, faster in warm, moist soil and fastest when finely divided; in a study of 10 Australian cropping soils, it oxidised faster as soil pH rose (Zhao et al. 2015).	As general soil chemistry, spoken. Never as a rate or a day count for this product.
The diagnostic (general agronomy)	Sulfur shortage shows on the youngest leaves and persists after nitrogen is applied; nitrogen shortage shows on the oldest leaves.	As the field test a dealer teaches.
Not on VAN's record	A Green Sulfur trial; a yield figure; a Pakistani sulfur deficiency percentage; a particle size or an oxidation rate; a foliar or drone rate.	Not said.

Sources: the Green Sulfur specification sheet and label; the regulatory table on van.com.pk; the Punjab soil survey (770,160 samples, tested in about 2016 to 2018) as van.com.pk publishes it; Zhao, Degryse, Gupta and McLaughlin 2015; the youngest-leaf diagnostic is general agronomy.

On wheat, Green Sulfur costs Rs 1,900 per acre for 1.4 kg of sulfur; on maize, Rs 9,500 for 7 kg.

Prices are the delivered farmer price from the VAN farmer price sheet, August 2026. Sulfur quantities are calculated from the declared analysis, 70% elemental sulfur.

Green Sulfur, S 70%	Weight	Sulfur	Applications	Rupees
1 pack of 1 kg	1 kg	0.7 kg		950
1 pack of 5 kg	5 kg	3.5 kg		4,250
1 bag of 20 kg	20 kg	14 kg		15,750
Wheat: 1 kg at germination, 1 kg at early growth, in 1 kg packs	**2 kg**	**1.4 kg**	**2**	**1,900**
Maize: 5 kg at land preparation, 5 kg at germination, in 1 kg packs	**10 kg**	**7 kg**	**2**	**9,500**

On wheat, the grower spends Rs 1,900 per acre to put 1.4 kg of sulfur through the water at the 2 stages the crop plan names. On maize, he spends Rs 9,500 per acre for 10 kg of product in 1 kg packs, 5 kg at land preparation and 5 kg at germination. The 10 kg holds 7 kg of sulfur. VAN publishes no yield or crop-response figure for Green Sulfur.

Prices: the VAN farmer price sheet, August 2026. 0.7 kg = 1 × 0.70; 14 kg = 20 × 0.70. Rupees: kg × 950 for 1 kg packs; kg ÷ 20 × 15,750 for the 20 kg bag. Rates: VAN's published crop plans on van.com.pk.

Green Sulfur per acre in the crop plans: the field crops.

Packs per acre are from the published crop plans. Sulfur is calculated from the declared analysis, and rupees per acre from the delivered farmer prices in the VAN farmer price sheet, August 2026, for the pack each row names.

Crop	Green Sulfur per acre, as printed	Sulfur	Rupees	Stage and method, as printed
Wheat	1 kg, 2 times, in 1 kg packs	1.4 kg	1,900	Germination, fertigation; Early Growth, fertigation
Maize	5 kg, 2 times, in 1 kg packs	7 kg	9,500	Land Preparation, drill at sowing, broadcast if no drill; Germination, fertigation
Cotton	5 kg, 2 times, in 1 kg packs	7 kg	9,500	Land Preparation, fertigation; Grand Growth, fertigation
Sugarcane	5 kg, 2 times, in 1 kg packs	7 kg	9,500	Land Preparation, drill at sowing, broadcast if no drill; Early Growth, fertigation
Basmati Rice	2 kg, 2 times, in 1 kg packs	2.8 kg	3,800	Establishment, fertigation; Grand Growth, fertigation
Canola	5 + 1.5 + 1.5 kg, in 1 kg packs	5.6 kg	7,600	Land Preparation, drill at sowing, broadcast if no drill; Germination, side dressing/broadcasting; Early Growth, fertigation
Sunflower	5 + 2.5 + 2.5 kg, in 1 kg packs	7 kg	9,500	Land Preparation, drill at sowing, broadcast if no drill; Germination, side dressing/broadcasting; Early Growth, fertigation

Most field crops get 5.6 kg to 7 kg of sulfur per acre in 2 or 3 applications; wheat and rice get less. The method is in the last column.

Rows as printed in VAN's published crop plans on van.com.pk; rupees for the pack the crop plan names (1 kg at Rs 950, 20 kg at Rs 15,750). Sulfur = kg × 0.70.

Green Sulfur per acre in the crop plans: the vegetables.

Packs per acre are from the published crop plans. Sulfur is calculated from the declared analysis, and rupees per acre from the delivered farmer prices in the VAN farmer price sheet, August 2026, for the pack each row names.

Crop	Green Sulfur per acre, as printed	Sulfur	Rupees	Stage and method, as printed
Potato	10 kg, from the 20 kg bag	7 kg	7,875	Land Preparation, drill at sowing, broadcast if no drill
Garlic	10 kg, from the 20 kg bag	7 kg	7,875	Land Preparation, drill at sowing, broadcast if no drill
Onion	2 kg, 3 times, in 1 kg packs	4.2 kg	5,700	Early Growth (20 DAT), fertigation; Grand Growth (40 DAT), fertigation; Bulb filling (50 DAT), fertigation
Chili	2 kg, 2 times, in 1 kg packs	2.8 kg	3,800	Early Growth, fertigation; Grand Growth, fertigation

The crop plan names 1 pack for each crop: 1 kg packs where every dose is 5 kg or less, and the 20 kg bag where a dose is larger, as on potato and garlic. The method column is copied from the crop plan; for soil applications, follow the order on the method slides: drill first, band second, broadcast last. DAT means days after transplanting.

Rows as printed in VAN's published crop plans on van.com.pk; rupees for the pack the row names. Sulfur = kg × 0.70.

An acre of wheat at 40 maunds, with its straw, takes out an estimated 6.4 kg of sulfur; the crop plan puts on 4.84 kg.

Estimated sulfur taken out of 1 acre by a wheat crop of 40 maunds (1.6 tonnes), calculated from the published removal tables used on van.com.pk, compared with the sulfur from the 3 products in the wheat crop plan.

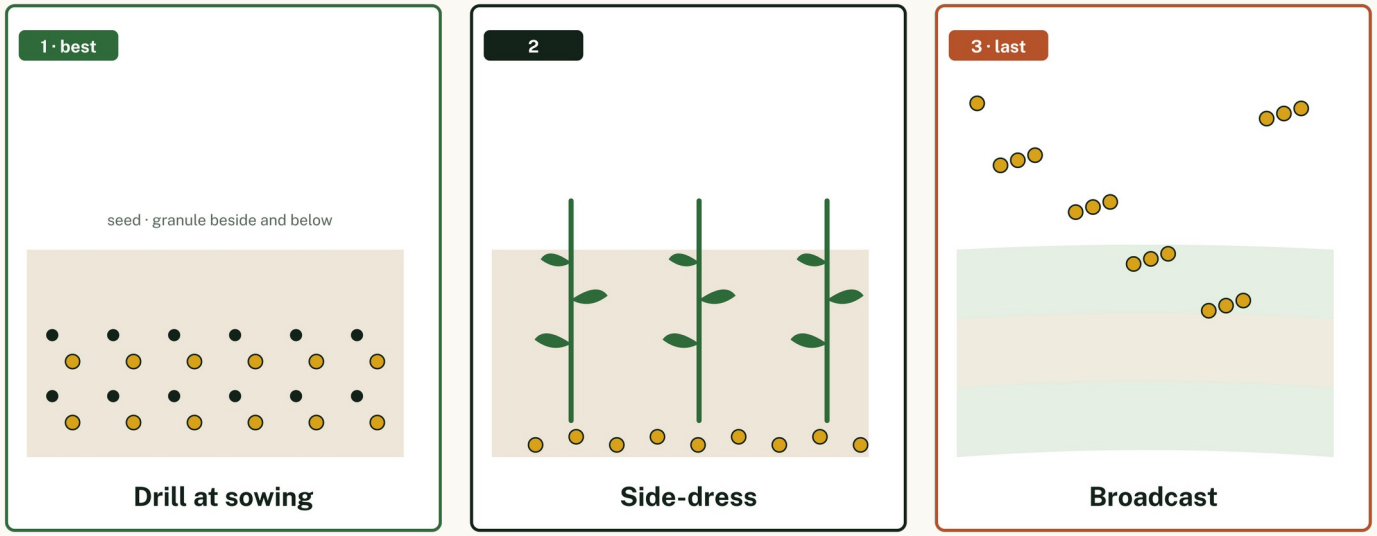
Wheat, 1 acre at 40 maunds	Sulfur per tonne of grain	Sulfur at 1.6 t	
Grain taken out	1.7 kg	2.7 kg	
Straw (bhusa) taken out too, per tonne of grain	2.3 kg	3.7 kg	
Grain and straw together	**4.0 kg**	**6.4 kg**	
Green Sulfur, 2 packs of 1 kg, puts on		**1.4 kg**	
Vital Urea, 1 bag of 25 kg at S 13%, puts on		**3.25 kg**	
Fusion Potash, ½ bag of 25 kg at S 1.5%, puts on		**188 g**	
The 3 lines together		**4.84 kg**	

Where the bhusa is also removed, the crop takes out an estimated 6.4 kg of sulfur per acre, and the crop plan puts on 4.84 kg from 3 products. The rest of the sulfur must come from the soil. How much sulfur Punjab's soils hold is not known, because the samples in the Punjab soil survey were not tested for sulfur. These figures show why sulfur is applied; they are not a yield claim.

Removal per tonne: the nutrient removal tables of the International Plant Nutrition Institute (IPNI), metric, 01/15: wheat grain 1.7 kg S and straw 2.3 kg S per tonne of grain, as van.com.pk uses them. 40 maunds × 40 kg = 1.6 t. Green Sulfur 2 × 1 kg × 0.70 = 1.4 kg. Vital Urea 1 bag = 25 kg × 0.13 = 3.25 kg; Fusion Potash ½ bag = 12.5 kg × 0.015 = 188 g (the wheat crop plan). Soil: 0 of the 770,160 Punjab soil survey samples tested for sulfur.

HOW IT IS APPLIED

Drill or broadcast the granule at sowing; put the powder through the water where the row says fertigation.



1. Drill at sowing, best

The seed drill's fertiliser box places the granule about 5 cm beside and below the seed row, never touching the seed. The microbes are in the moist root zone, so that is where the granule should be. Every land-preparation soil row in the crop plans says: drill at sowing, or broadcast if there is no drill.

2. Side-dress, second

A band beside the row, by hand or planter, at sowing or the first irrigation, covered with soil. For canola and sunflower, the crop plan row at germination says side dressing or broadcasting.

3. Broadcast, last; fertigation where the row prints it

Spread it before the last ploughing or before an irrigation, and mix it in. Granules left on the surface are away from the microbes and the root. Where the row says fertigation (wheat, rice, cotton, onion, most later stages), use the fertigation powder, which disperses in the drum and goes through the water at that stage. Apply it before the crop needs it, because oxidation takes time.

Drilling is best, side-dressing next, broadcasting last.

VAN gives the same placement order for every granule it sells for the soil at sowing, including the Green Sulfur granule. The other grade, the fertigation powder, is for the rows that say fertigation.

Preference	Method	What it means	Why
1 · best	Drill at sowing	The seed drill's fertiliser box places the granule about 5 cm beside and below the seed row, never touching the seed.	The root reaches a drilled granule sooner than a side-dressed or broadcast one. The granule has the least contact with air, sun and free calcium at the surface, and all of the fertiliser is placed in the root zone.
2 · second best	Side-dress	The granules are placed in a band beside the row, by hand or with a planter, at sowing or at the first irrigation, and covered with soil.	The band is still near the root, but it goes on later and shallower than a drilled granule.
3 · last option	Broadcast	The granules are spread on the surface before the last ploughing or before an irrigation, and mixed in.	Most of the fertiliser lies away from the root, and most of it stays near the surface, in contact with air, sun and free calcium.

In VAN's own fields and partner fields, on wheat, maize and sugarcane, basal fertiliser drilled at sowing has given 7 to 9% more yield per acre than the same bag broadcast: the mean of replicated observations across regions, not a published trial.

This order applies where Green Sulfur granules go on the soil. Where the crop plan row says fertigation, follow that row.

OBJECTION 1 OF 8

“15,750 for a 20 kg bag of sulfur? Sulfur is cheap.”

What the dealer says

Write the price down, and then write what the bag holds: 14 kg of elemental sulfur made into a granule that breaks into fine particles in water. You spread it as a granule, and the soil microbes work on it as fine particles. Wheat takes 2 kg, Rs 1,900 per acre, at the 2 stages the crop plan names. Maize takes 10 kg over the season. You are paying for sulfur in a form you can spread, which stays in the soil until the microbes turn it into sulfate, a little at a time.

The facts behind the answer

- 1 kg Rs 950; 5 kg Rs 4,250; 20 kg Rs 15,750, delivered (the farmer price sheet, August 2026).
- Wheat 2 kg per acre, Rs 1,900; maize 10 kg over the season in 1 kg packs, Rs 9,500 (the crop plans).
- 14 kg of sulfur in a 20 kg bag; 0.7 kg in a 1 kg pack (the declared analysis).
- The granule (WDG): 70% sulfur with a wetting and a dispersing agent (the specification sheet).

Never say: “It gives you X more maunds” or a rupees-per-acre return. No yield figure exists for the product. Do not compare the price per kg of sulfur with another product.

OBJECTION 2 OF 8

“My crop is yellow now. Will this green it up this week?”

What the dealer says

No, and VAN does not sell it to rescue a crop. The plant cannot use elemental sulfur on the day it is applied. It becomes available only when soil microbes turn it into sulfate. That takes time, though less in warm, moist soil. If the crop is short of sulfur today, a sulfate source is the answer. Green Sulfur supplies sulfate over the season: put it on at the crop plan stage, before the crop needs it, and sulfate forms as the crop grows. First check the young leaves, so you know the shortage is sulfur and not nitrogen.

The facts behind the answer

- Elemental sulfur is not available to the plant as applied; soil microbes oxidise it to sulfate over time (general soil chemistry).
- If the crop needs sulfur now, use a sulfate source; elemental sulfur supplies sulfate over the season.
- Sulfur shortage shows on the young leaves, all over; nitrogen shortage starts on the old leaves (the 17-nutrients page on van.com.pk).

Never say: “It works in 3 days” or any day count. VAN prints no oxidation rate for the product.

OBJECTION 3 OF 8

“I already put sulfur on with the urea. Vital Urea has 13%.”

What the dealer says

Yes, and the crop plan counts it. On wheat, the crop plan puts on 4.84 kg of sulfur per acre: 3.25 kg from 1 bag of Vital Urea, 1.4 kg from 2 packs of Green Sulfur and 188 g from ½ bag of Fusion Potash. A wheat crop of 40 maunds per acre is estimated to take out about 6.4 kg of sulfur in the grain and bhusa together. Vital Urea and Green Sulfur are on the same crop plan on purpose: Vital Urea supplies sulfur with the nitrogen at the nitrogen stages, and Green Sulfur supplies it at its own stages. Neither replaces the other. Your soil report cannot tell you whether your soil is short of sulfur, because sulfur was not tested.

The facts behind the answer

- The wheat crop plan: Vital Urea 1 bag (S 13%), Fusion Potash ½ bag (S 1.5%), and Green Sulfur 1 kg at germination and again at early growth (the crop plan on van.com.pk).
- A wheat crop removes 1.7 kg of sulfur in each tonne of grain, and a further 2.3 kg in the straw that comes with that tonne (IPNI removal tables).
- 0 of the 770,160 samples in the Punjab soil survey (about 2016 to 2018) were tested for sulfur.
- Not every field is short of sulfur (Ahmad et al. 2003), so a dealer never says that every field is.

Never say: “Every field is short of sulfur.”

OBJECTION 4 OF 8

“Can I put it through the drip or the flood water?”

What the dealer says

Yes, where the crop plan row says fertigation, and most rows do: wheat at germination and early growth, rice, cotton at land preparation and grand growth, ratoon sugarcane, and onion 3 times. Ask for the fertigation powder; the granule is for the soil. Stir the powder into the drum and release it across the whole irrigation, not all at once. Where the land-preparation row says drill at sowing, on maize, February-planted sugarcane, canola, sunflower, potato and garlic, the granule goes on the soil: drilled at sowing beside and below the seed row, or broadcast and worked in if there is no drill.

The facts behind the answer

- 2 grades: the granule (WDG) for the soil and the fertigation powder for the water (VAN, 27 Sep 2026).
- The label names 2 routes: soil and fertigation.
- Of the 42 Green Sulfur rows in the 22 crop plans, 32 are fertigation, 8 are land preparation (drill at sowing, or broadcast) and 2 are side dressing or broadcasting (the crop plans on van.com.pk).
- The granule is made to be spread dry and to break up once it is wet in the soil; for the drip, use the powder (the specification sheet; van.com.pk).

Never say or give: any foliar rate or drone rate. Neither is on any crop plan row.

OBJECTION 5 OF 8

“My soil is alkaline. Will sulfur fix the pH?”

What the dealer says

Not across the field, and VAN does not say it will. While the microbes turn a particle into sulfate, the soil within a few millimetres of it becomes acidic, and phosphate and zinc held by the lime can come back into solution there. The acid effect stays within a few millimetres of each particle for the weeks the conversion takes. A soil test measures the whole field, so it will not show this effect. At the crop plan rates, 2 to 10 kg of Green Sulfur per acre, the pH on the soil report will not change. The change is only in the soil next to each particle, where more phosphate and zinc can come back into solution for the root.

The facts behind the answer

- Sulfur releases acid only into the soil within a few millimetres of it; the effect is local, not across the field (the soil register on van.com.pk).
- Elemental sulfur makes the soil around each particle acidic, which can bring phosphate and zinc held by lime back into solution in that small zone (the 17-nutrients page on van.com.pk).
- 69.1% of 770,160 samples tested above pH 8; the mean pH was 8.22 (the Punjab soil survey, about 2016 to 2018).

Never say: “It reclaims sodic soil” or “it brings your pH down”.

OBJECTION 6 OF 8

“Which stage, and how many packs?”

What the dealer says

Use what the crop plan prints for your crop. Wheat: 1 kg at germination, 1 kg at early growth, in 1 kg packs. Maize: 5 kg at land preparation, 5 kg at germination, in 1 kg packs. Cotton: 5 kg at land preparation, 5 kg at grand growth, in 1 kg packs. Potato: 10 kg at land preparation, from the 20 kg bag. Onion: 2 kg at early growth (20 DAT), 2 kg at grand growth (40 DAT), 2 kg at bulb filling (50 DAT), in 1 kg packs. Always apply it before the stage that needs it, because the microbes need time.

The facts behind the answer

- Rates per acre as printed in the crop plans on van.com.pk; 12 crops on the crop plan slides.
- Apply it before the crop needs sulfur, because the microbes take time to turn it into sulfate (general soil chemistry).
- Sulfate beyond the crop’s need leaches like nitrate, so the sulfate formed from sulfur applied above the crop plan rate can be washed below the root.

Never say: “Double it if the crop is very pale.”

OBJECTION 7 OF 8

“Is it registered? What is the licence?”

What the dealer says

It is registered under Pakistan Standard PS 5449-2020, with manufacturing licence CM/L-2962/2023, and VAN is the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Enter the batch number from the bag on van.com.pk, and you can read the certificate for that batch.

The facts behind the answer

- Green Sulfur: PS 5449-2020, CM/L-2962/2023 (the regulatory table on van.com.pk).
- The licence number is what a dealer quotes; no expiry date is printed.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017.

Never say or give: a licence expiry date, or that the wetting and dispersing agents are nutrients.

OBJECTION 8 OF 8

“Is it a fungicide? Sulfur is sprayed for mildew.”

What the dealer says

Sulfur dust and wettable sulfur are sprayed as fungicides. They are different products, used at different rates and under a different licence. Green Sulfur is registered as a fertiliser. It is applied to the soil or through the water at the crop plan rate, and VAN makes no disease claim for it. A grower with a mildew problem needs a registered fungicide and a crop-protection dealer.

The facts behind the answer

- Registered under PS 5449-2020 as a fertiliser (the regulatory table on van.com.pk).
- No crop plan row prints a foliar use for Green Sulfur.
- VAN makes no disease or immunity claim for any product.

Never say: “It controls mildew too” or any disease or immunity effect.

Green Sulfur in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Germination	1 kg	Fertigation	1 kg
Wheat	Early Growth	1 kg	Fertigation	1 kg
Maize (Corn)	Land Preparation	5 kg	Drill at sowing; broadcast if no drill	1 kg
Maize (Corn)	Germination	5 kg	Fertigation	1 kg
Basmati Rice	Establishment	2 kg	Fertigation	1 kg
Basmati Rice	Grand Growth	2 kg	Fertigation	1 kg
Hybrid Rice	Early Growth	2 kg	Fertigation	1 kg

The rows are printed as published in the crop plans on van.com.pk. Coverage: these 4 slides show 12 of the 22 crop plans that name Green Sulfur, and 25 of its 42 rows. Of these 25 rows, 17 say fertigation, 6 say drilling at sowing (or broadcasting) and 2 say side-dressing or broadcasting. The rows use the 1 kg pack and the 20 kg bag. Each rate is per acre, and the crop page scales it to the size of the field.

Green Sulfur in the published crop plans, continued.

Crop	Stage	Rate per acre	Method	Pack
Hybrid Rice	Grand Growth	2 kg	Fertigation	1 kg
Cotton	Land Preparation	5 kg	Fertigation	1 kg
Cotton	Grand Growth	5 kg	Fertigation	1 kg
Sugarcane, February planting	Land Preparation	5 kg	Drill at sowing; broadcast if no drill	1 kg
Sugarcane, February planting	Early Growth	5 kg	Fertigation	1 kg
Ratoon Sugarcane	Land Preparation	5 kg	Fertigation	1 kg
Ratoon Sugarcane	Early Growth	5 kg in April	Fertigation	1 kg

The rows continue from the previous slide. Each rate is per acre, and the crop page scales it to the size of the field.

Green Sulfur in the published crop plans, continued.

Crop	Stage	Rate per acre	Method	Pack
Canola	Land Preparation	5 kg	Drill at sowing; broadcast if no drill	1 kg
Canola	Germination	1.5 kg	Side dressing/Broadcasting	1 kg
Canola	Early Growth	1.5 kg	Fertigation	1 kg
Sunflower	Land Preparation	5 kg	Drill at sowing; broadcast if no drill	1 kg
Sunflower	Germination	2.5 kg	Side dressing/Broadcasting	1 kg
Sunflower	Early Growth	2.5 kg	Fertigation	1 kg
Potato	Land Preparation	10 kg	Drill at sowing; broadcast if no drill	20 kg

The rows continue from the previous slide. Each rate is per acre, and the crop page scales it to the size of the field.

Green Sulfur in the published crop plans, continued.

Crop	Stage	Rate per acre	Method	Pack
Onion	Early Growth (20 DAT)	2 kg	Fertigation	1 kg
Onion	Grand Growth (40 DAT)	2 kg	Fertigation	1 kg
Onion	Bulb filling (50 DAT)	2 kg	Fertigation	1 kg
Garlic	Land Preparation	10 kg	Drill at sowing; broadcast if no drill	20 kg

The rows continue from the previous slide. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield figure, or that it greens a crop in X days. No Green Sulfur trial exists; the release of sulfate over the season is described from how sulfur works in soil, and no oxidation rate or day count is given.

2. That every field is short of sulfur, or “88% of soils are deficient”. None of the 770,160 samples in the Punjab soil survey was tested for sulfur, so the survey says nothing about sulfur levels. Ahmad et al. 2003 found fields that were not short of sulfur.

3. A soil pH change, a sodic-soil reclamation, or a soil-test change at the crop plan rate. The acidic effect stays within a few millimetres of each particle while it oxidises.

4. A disease, mildew or immunity effect. It is registered as a fertiliser, and no disease claim is made.

5. A foliar or drone rate. No crop plan row prints foliar for Green Sulfur, and no drone grade is in this deck.

6. A ratio of nitrogen saved per kilogram of sulfur, or that it “unlocks” soil phosphorus and zinc across the field. VAN claims only the effect in the small zone around each particle.

HOW A GROWER VERIFIES A BAG

Every batch is tested, and the batch number on the bag identifies its certificate.

1 · The batch number

Every bag carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any bag by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for Green Sulfur shows

The certificate shows the tested elemental sulfur content of the batch against the declared minimum of 70%. It also shows the batch number, and the Pakistan Standard PS 5449-2020 and licence CM/L-2962/2023 printed on the bag. No particle-size or oxidation-rate figure is on the bag or in this training. This training shows no batch result; a grower can read the certificate of his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEWED 27 SEPTEMBER 2026

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Sulfur turns the nitrogen you applied into protein. *English meaning · Urdu reviewed 27 Sep 2026*	سلفر آپ کی دی ہوئی نائٹروجن کو پروٹین میں بدلتا ہے
If the new leaves are pale and urea did nothing, the crop is short of sulfur. *English meaning · Urdu reviewed 27 Sep 2026*	اگر نئے پتے پیلے ہیں اور یوریا نے کچھ نہ کیا، تو کمی سلفر کی ہے
Put it on before the crop needs it. The soil takes time to turn it into food. *English meaning · Urdu reviewed 27 Sep 2026*	فصل کی ضرورت سے پہلے دیں، زمین کو اسے خوراک بنانے میں وقت لگتا ہے
Dry, it is a granule. In water, it becomes a fine powder. *English meaning · Urdu reviewed 27 Sep 2026*	خشک حالت میں دانہ ہے، پانی میں باریک پاؤڈر بن جاتا ہے

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

Green Sulfur is drilled at sowing, broadcast and worked in, or put through the water where the row says, always before the crop needs it.

